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Department of Computer Science
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F.Y.B.Sc.(Data Science) Sem – II
DS-152-P: Lab Course on DS-151-T (Advanced Python Programming)

Practical Assignment5

Date: 17/01/2025 and 24/01/2025

1. Write a Python GUI program to import Tkinter package and create a window. Set its title and add a label to the window.

```
from tkinter import *  
window=Tk()  
window.title("Tkinter Label Demo")  
window.geometry("500x500")  
l1=Label(window, text="My First Tkinter Label", bg="Yellow", fg="Red")  
l1.pack()  
window.mainloop()
```

2. Write a Python GUI program to create a label and change the label font style (font name, bold, size) using tkinter module.

```
from tkinter import *  
window=Tk()  
window.title("Tkinter Label")  
window.geometry("500x500")  
l1=Label(window, text="My First Tkinter Label", bg="Yellow", fg="Red",  
font=("Arial", 16, "bold"))  
l1.pack()  
window.mainloop()
```

3. Create a Tkinter window with multiple labels displaying different text messages and styled with different fonts and colors.

```
from tkinter import *
```

```
window=Tk()
```

```
window.title("Tkinter Label")
```

```
window.geometry("500x500")
```

```
l1=Label(window, text="My First Tkinter Label", bg="Pink", fg="Red",  
font=("Arial", 16, "bold"))
```

```
l1.pack(side = LEFT)
```

```
l2=Label(window, text="My second Tkinter Label", bg="Pink", fg="Green",  
font=("Times New Roman", 18, "italic"))
```

```
l2.pack(side = RIGHT)
```

```
l3=Label(window, text="My Third Tkinter Label", bg="Yellow", fg="Blue",  
font=("Monotype Corsiva", 20, "bold"))
```

```
l3.pack(side = TOP)
```

```
l4=Label(window, text="My Fourth Tkinter Label", bg="yellow", fg="Pink",  
font=("Cambria", 22, "bold"))
```

```
l4.pack(side = BOTTOM)
```

```
window.mainloop()
```

4. Create a Tkinter window with a button. When the button is clicked, display a message in a label.

```
from tkinter import *
top = Tk()
top.geometry("200x200")
def helloCallBack():
    l=Label(text="Hello World")
    l.place(x=100,y=100)
B = Button(top, text ="Hello", command = helloCallBack)
B.place(x=50,y=50)
top.mainloop()
```

5. Create a Tkinter window with a button. When the button is clicked, display a message using messagebox.

```
from tkinter import *
from tkinter import messagebox
top = Tk()
top.geometry("200x200")
def helloCallBack():
    msg=messagebox.showinfo( "Hello Python", "Hello World")

B = Button(top, text ="Hello", command = helloCallBack)
B.place(x=50,y=50)
top.mainloop()
```

6. Create a Tkinter window with two frames. The first frame should contain a label and the second frame should contain two buttons side by side.

```
from tkinter import *  
window= Tk()  
window.geometry("700x350")  
  
f1=Frame(window, relief="sunken", bg="Pink", borderwidth=2, width=200,  
height=250)  
f1.pack(side="left", fill="y")  
  
l1=Label(f1, text="My First Tkinter Label",bg="Yellow",fg="Red")  
l1.pack(side="left",padx=20,pady=20)  
  
f2=Frame(window, relief="sunken", bg="skyblue3", borderwidth=2,width=400,  
height=250)  
f2.pack(side="left", fill= "y")  
  
b1=Button(f2, text="My First Button",bg="Yellow",fg="Red")  
b1.pack(side="left",padx=20,pady=20)  
  
b2=Button(f2, text="My Second Button",bg="Yellow",fg="Red")  
b2.pack(side="left",padx=20,pady=20)  
  
window.mainloop()
```

7. Create a Tkinter window with a frame containing three buttons. Arrange the buttons vertically within the frame.

```
from tkinter import *
```

```
window= Tk()
```

```
window.geometry("700x350")
```

```
f1=Frame(window, relief="sunken", bg="Pink", borderwidth=2, width=200,  
height=250)
```

```
f1.pack(padx=20,fill="y")
```

```
b1=Button(f1, text="My First Button",bg="Yellow",fg="Red")
```

```
b1.pack(padx=20,pady=20)
```

```
b2=Button(f1, text="My Second Button",bg="Yellow",fg="Red")
```

```
b2.pack(padx=20,pady=20)
```

```
b3=Button(f1, text="My Third Button",bg="Yellow",fg="Red")
```

```
b3.pack(padx=20,pady=20)
```

```
window.mainloop()
```

8. Create a Tkinter window with multiple labels and place them using grid.

```
from tkinter import *  
window=Tk()  
window.title("Tkinter Label")  
window.geometry("500x500")  
  
l1=Label(window, text="My First Tkinter Label", bg="Yellow", fg="Red",  
font=("Arial", 16, "bold"))  
l1.grid(row=0,column=0)  
  
l2=Label(window, text="My second Tkinter Label", bg="Yellow", fg="Red",  
font=("Arial", 16, "bold"))  
l2.grid(row=1,column=5)  
  
window.mainloop()
```

9. Write a python program for addition of 2 numbers using GUI (Use tkinter – Label, Entry, Button)

```
from tkinter import *  
from tkinter import messagebox  
  
window=Tk()  
  
window.geometry("500x500")  
Label(window,text="Sum of 2 No",font="arial 14 bold")  
  
v1=StringVar()  
v2=StringVar()  
v3=StringVar()
```

```
l1=Label(window,text="Enter Number1: ")
l1.place(x=10,y=10)
e1=Entry(window,textvariable=v1)
e1.place(x=100,y=10)
```

```
l2=Label(window,text="Enter Number2: ")
l2.place(x=10,y=100)
e2=Entry(window,textvariable=v2)
e2.place(x=100,y=100)
```

```
l3=Label(window,text="result: ")
l3.place(x=10,y=200)
e3=Entry(window,textvariable=v3)
e3.place(x=100,y=200)
```

```
def sum():
    a=int(v1.get())
    b=int(v2.get())
    c=a+b
    v3.set(c)
```

```
def exit():
    window.destroy()
```

```
b1=Button(window,text="Sum",command=sum)
b1.place(x=30,y=300)
```

```
b2=Button(window,text="Exit",command=exit)
b2.place(x=100,y=300)
window.mainloop()
```